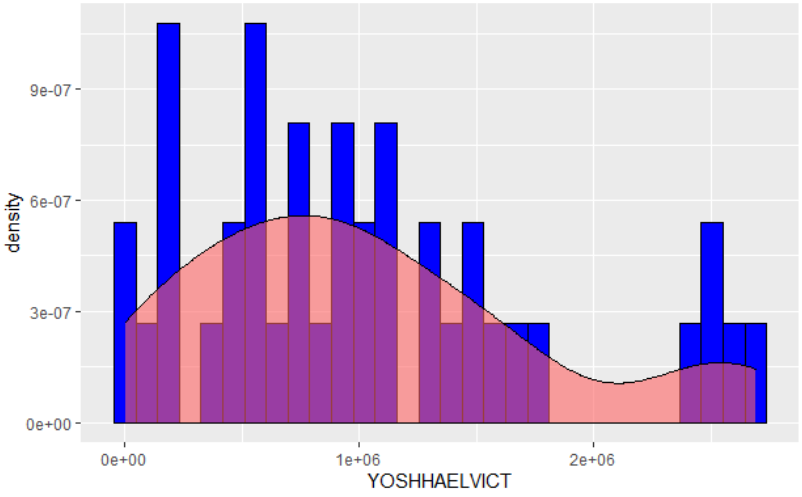
Exercise # 1:

The database (Sellin_Product.csv) comes from the database provided during the Fossil-Stock-Forecasting-Challenge machine learning competition. It gives the monthly sales quantity of three different products over the period 2016-01 to 2021-10. It has 4 columns and 58 rows. We can also notice the presence of some missing data. The three products are respectively named: "YOSHHAELVICT", "YOSHELLYCATH" and "YOSHLROYARTI".

We represent the price distributions of each of the products.

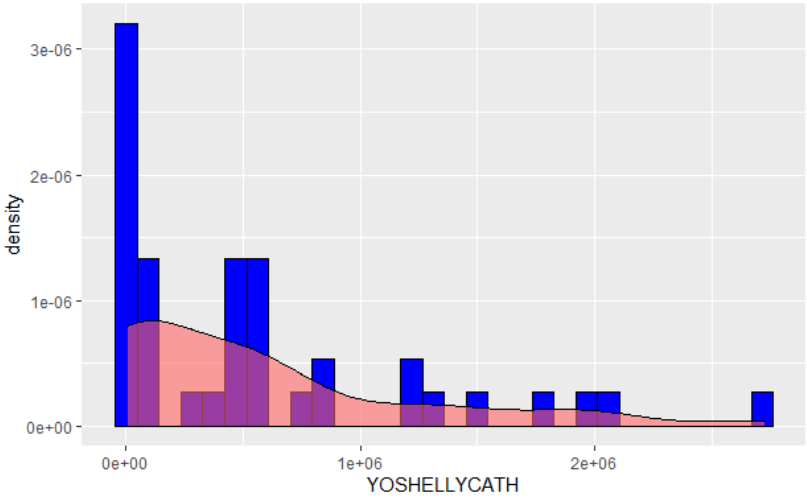
- ✓ YOSHHAELVICT

The average price of this product is 1029841 and the median is 938038.



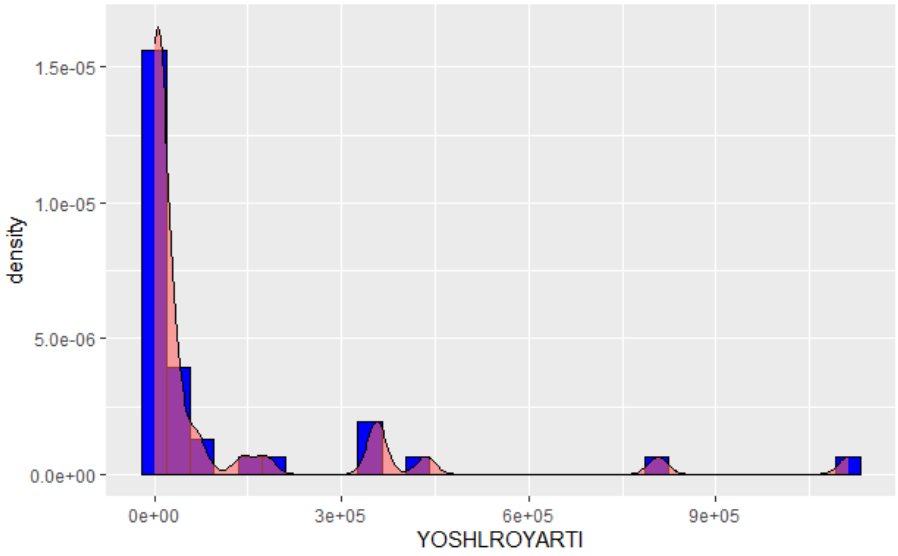
- ✓ YOSHELLYCATH

The average price of this product is 561733.8 and the median is 433564.



✓ YOSHLROYARTI

The average price of this product is 103756.5 and the median is 7091.



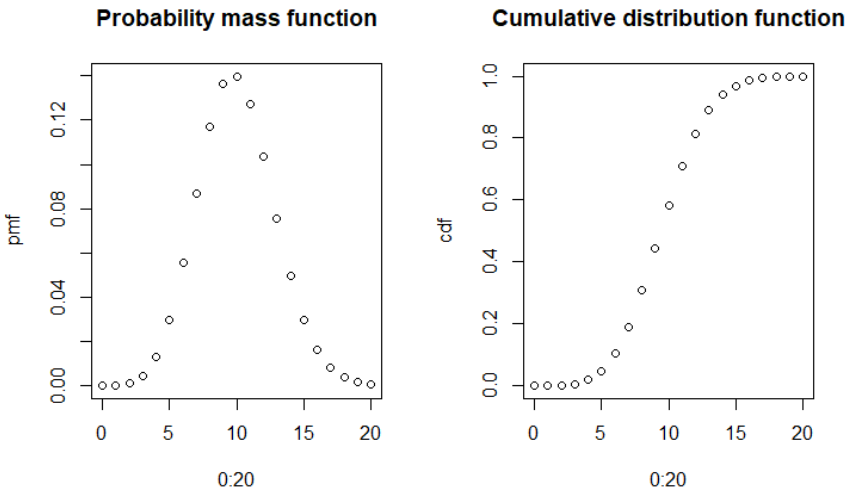
We notice that the distributions of the three variables are very different from each other.

Exercise # 2:

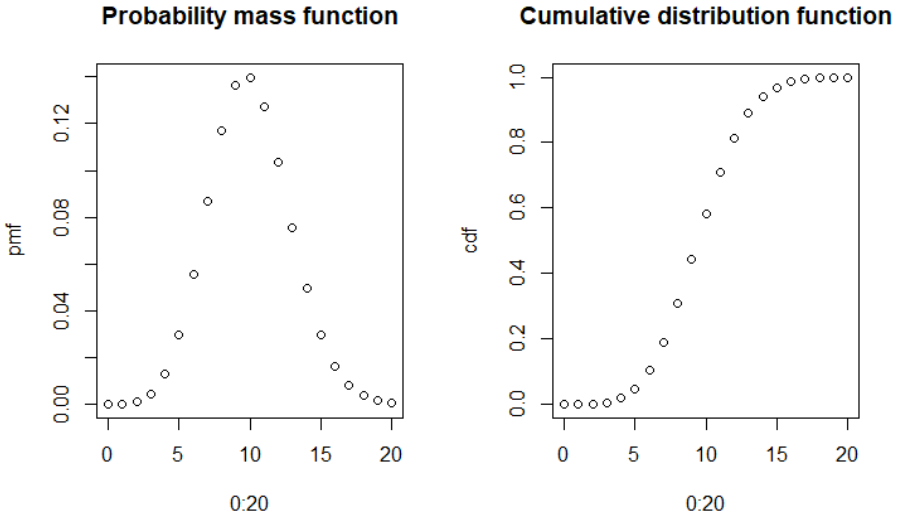
✓ **Binomial distribution**

We choose to use the following values: $p = \{0.2; 0.6\}$ and $n = \{50; 100\}$

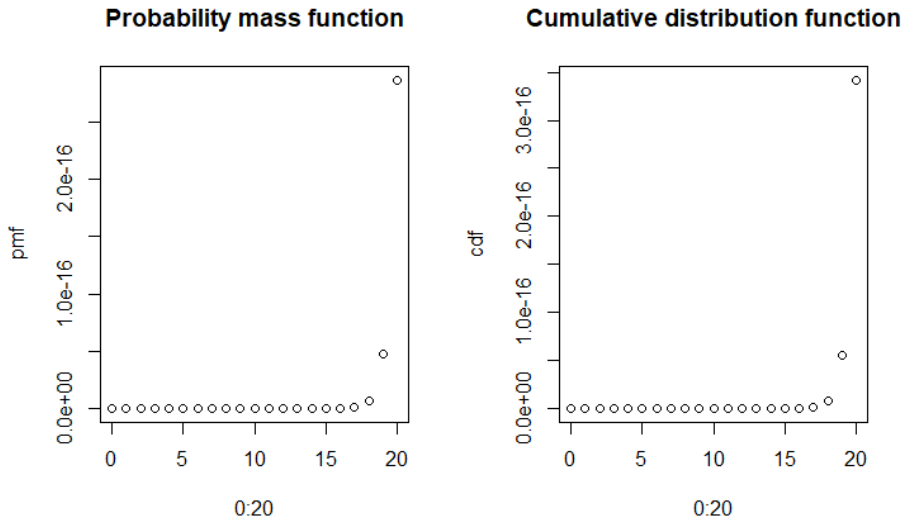
- For $p = 0.2$ and $n = 50$



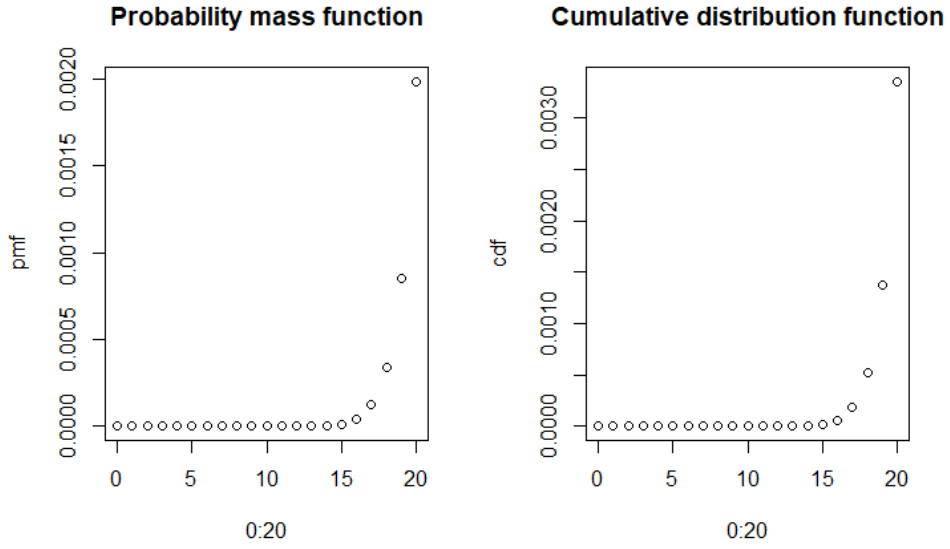
- For $p = 0.2$ and $n = 100$



- For $p = 0.6$ and $n = 100$



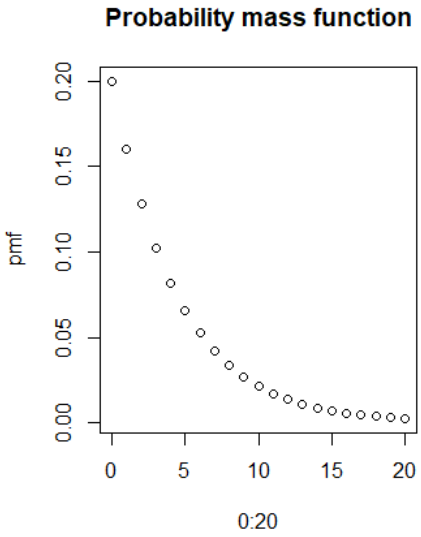
- For $p = 0.6$ and $n = 50$

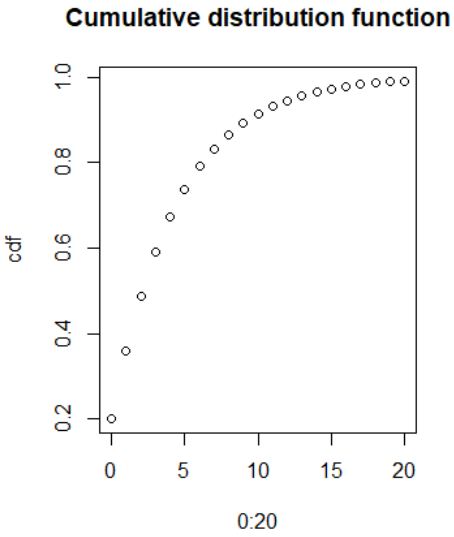


✓ <u>Geometric Distribution</u>

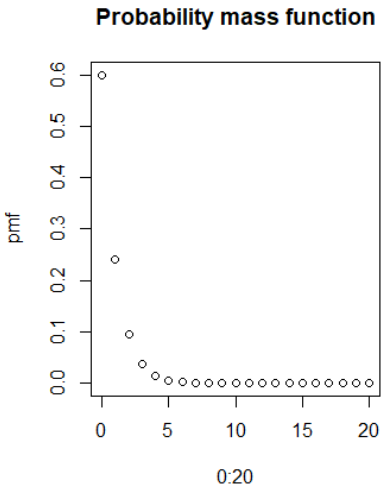
We choose to use the following values: $p1 = 0.2$ and $p2 = 0.6$

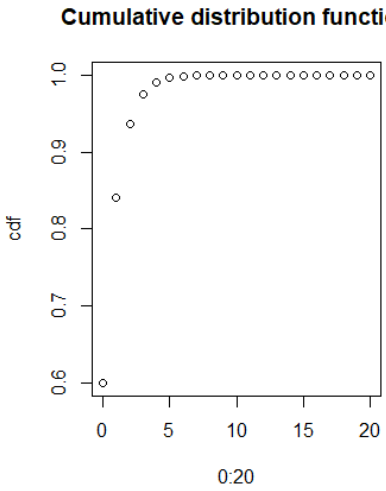
- $p1 = 0.2$





- $p2 = 0.6$

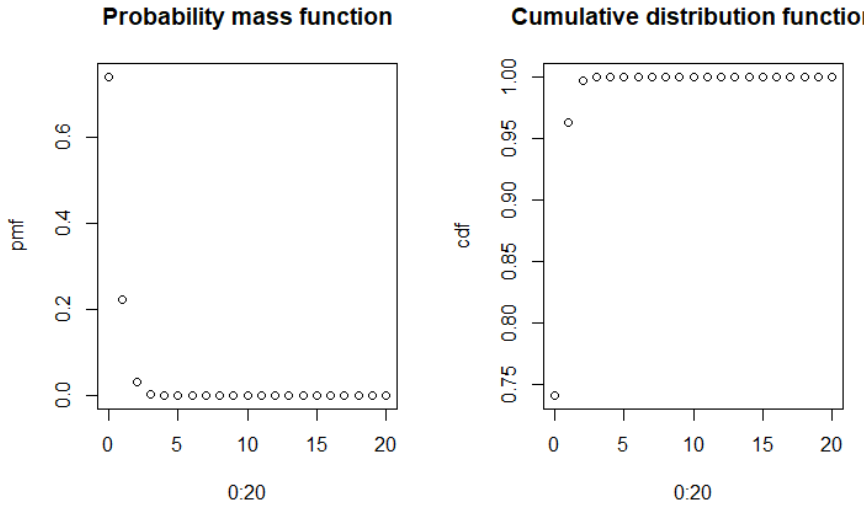




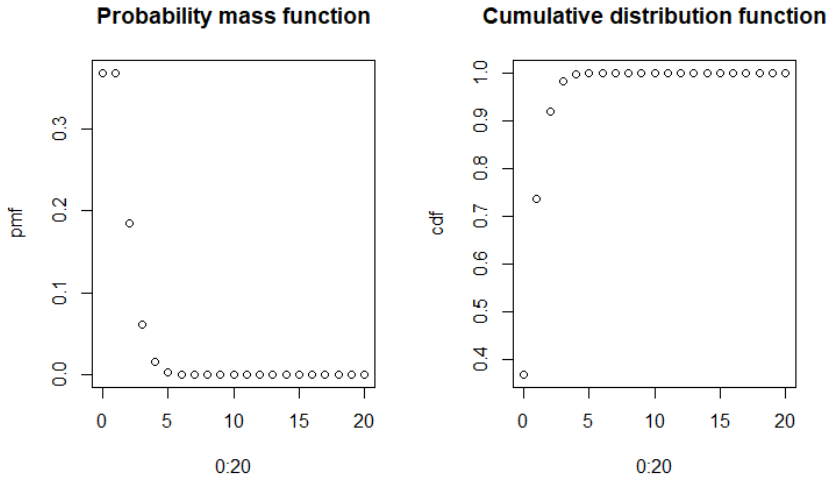
✓ **Poisson Distribution**

We choose to use the following values: $\lambda 1 = 0.3$ and $\lambda 2 = 1$

- $\lambda 1 = 0.3$



- $\lambda 2 = 1$



There is a big difference in the distributions. But we can note that the binomial distribution varies much more by modifying the parameter p which is the probability of success.

Notice:

This work was carried out using the RStudio software and the R programming language. The source codes of each exercise respectively named "Exercise_1.R" and "Exercise_2.R" are attached to this report.